When Embodied AI Meets Industry 5.0: Human-Centered Smart Manufacturing

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***Abstract*—As embodied intelligence (EI), large language models (LLMs), and cloud computing continue to advance, Industry 5.0 facilitates the development of industrial artificial intelligence (IndAI) through cyber-physical-social systems (CPSSs) with a human-centric focus. These technologies are organized by the system-wide approach of Industry 5.0, in order to empower the manufacturing industry to achieve broader societal goals of job creation, economic growth, and green production. This survey first provides a general framework of smart manufacturing in the context of Industry 5.0. Wherein, the embodied agents, like robots, sensors, and actuators, are the carriers for IndAI, facilitating the development of the self-learning intelligence in individual entities, the collaborative intelligence in production lines and factories (smart systems), and the swarm intelligence within industrial clusters (systems of smart systems). Through the framework of CPSSs, the key technologies and their possible applications for supporting the single-agent, multi-agent and swarm-agent embodied IndAI have been reviewed, such as the embodied perception, interaction, scheduling, multi-mode large language models, and collaborative training. Finally, to stimulate future research in this area, the open challenges and opportunities of applying Industry 5.0 to smart manufacturing are identified and discussed. The perspective of Industry 5.0-driven manufacturing industry aims to enhance operational productivity and efficiency by seamlessly integrating the virtual and physical worlds in a human-centered manner, thereby fostering an intelligent, sustainable, and resilient industrial landscape.**

***Index Terms*—Embodied AI, human-centered manufacturing, Industry 5.0, internet of things, large multi-mode language models.**

I. Introduction

IN year 2024, the researchers in the artificial intelligence (AI) domain have been honored with both the Nobel Prize in Chemistry and in Physics, marking a landmark event for the era of AI-driven advancements across various fields. The AI technology continues to support and enhance the manufacturing industry (MI), driving lean and agile production methods that boost flexibility, responsiveness, and efficiency [1]. In MI, AI helps to eliminate non-value-adding activities, achieving cost efficiency and high resource utilization in stable and predictable demand scenarios. Organizations can rapidly reconfigure their operations to meet shifting consumer demands and market conditions [2]. Industrial artificial intelligence (IndAI) is revolutionizing MI through advanced design and optimization [3], real-time monitoring [4], and human-robot collaboration [5]. For example, various object recognition models have been deployed in key industrial processes such as automated product assembly, unattended factory monitoring, and automated quality inspections. By leveraging embodied industrial entities — such as robotics, production lines, and workstations — IndAI can significantly enhance the intelligent capabilities of MI, leading to shorter, more efficient, and more flexible product cycles [6]. IndAI systems analyze data for proactive problem-solving and decision-making, offering several advantages: 1) the real-time insights for immediate, data-driven decision-making; 2) the flexible production scheduling to dynamically adjust and optimize resource use; 3) the quick customer response to rapidly adapt to changing demands; 4) the enhanced supply chain management for better coordination and visibility; and 5) the improved maintenance and reliability through predictive maintenance, reducing downtime and enhancing system performance.

In the advanced MI, the revolution of MI has been propelled by digital technologies, encompassing the entire lifecycle from product design through production, application, to service [7]. Nowadays, Industry 5.0 envisions a future of IndAI, which highlights its lean and agility in managing industrial production processes [8]. While Industry 4.0 emphasizes knowledge automation [9], data exchange [10], and internet of things (IoT) [11], Industry 5.0 marks a significant shift by focusing on a human-centric MI. The concept of Industry 5.0 was initially put forward by Wang in [9], centering at the parallel intelligence in cyber-physical-social systems (CPSSs). The CPSS refers to the enabling platform technology poised to usher in an era of intelligent enterprises and industries [12]. Such new paradigm aims to augment human capabilities and foster collaboration between humans and machines, rather than simply replacing human labor with automated systems [1]. Different from the tech-driven Indus-

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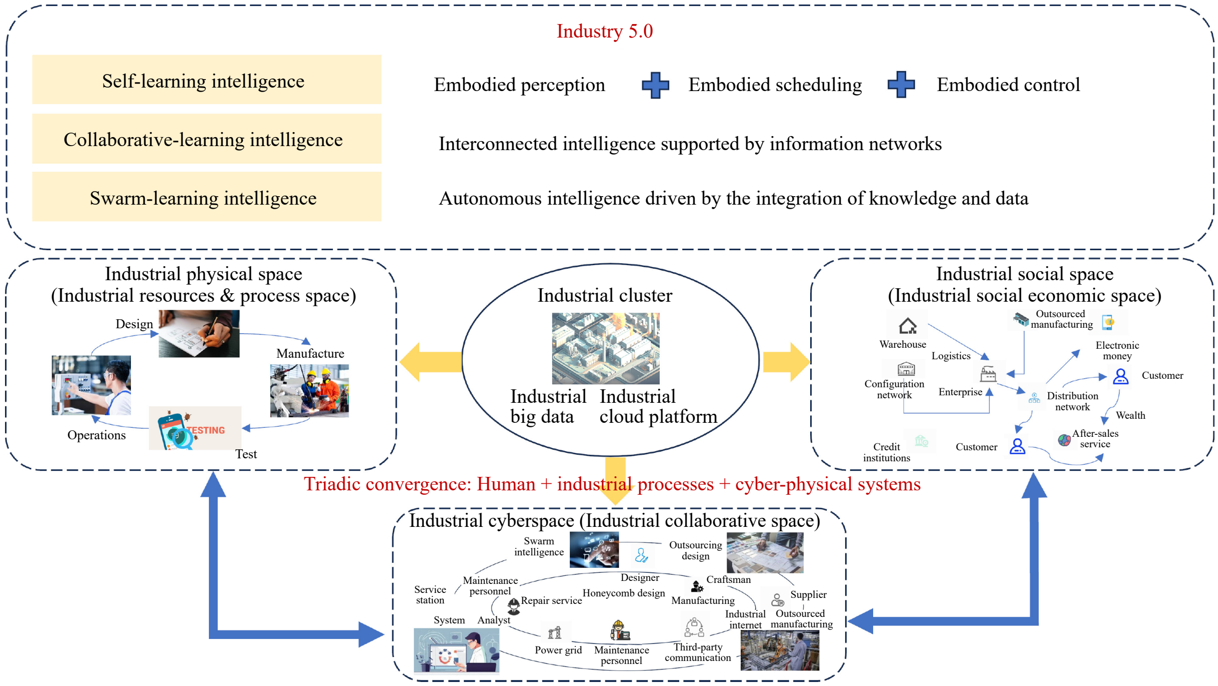

Fig. 1. The human-cyber-physical framework of smart MI in Industry 5.0.

try 4.0, Industry 5.0 is value-driven, centering around:

1) Human-centricity: The roles of human workers are clearly justified, which are essential decision-makers who supervise and interact with embodied machines [13].

2) Sustainability: The environmental and social impact of industrialization is required to advocate for sustainable developments that reuse and recycle natural resources, minimize waste, and lessen environmental impact [14].

3) Resilience: The capability of resilience involves ensuring that these CPSSs can withstand, recover from, and adapt to disruptions and crises in face of various challenges [8].

The synergistic relationship between humans and machines in Industry 5.0 represents a new paradigm that extends far beyond traditional manufacturing, which has been shown in Fig. 1. The boundary between human cognition and machine intelligence becomes increasingly blurred [15]. The goal shifts from the pure automation to the enhancement of human creativity and flexibility through the collaboration with intelligent machines [16]. Consequently, embodied entities are no longer merely tools for task execution; they are essential partners in dynamic and adaptive environments, facilitating flexibility, efficiency, and customization. These advancements push us closer to realize the concept of artificial general intelligence (AGI) [17], capable of generalizing across a broader range of tasks and applications in industrial contexts and beyond.

In Industry 5.0, each device, robot, production line, workshop, and factory can be regarded as an embodied entity. Such entities are interconnected and collaborated, forming a vast network of smart systems. The pursuit of creating smart and autonomous agents that can assist humans in industrial settings, and operate in real-world scenarios has been a foundational goal in AGI, contributing significantly to the advancement of Industry 5.0. The relationship between Industry 5.0 and embodied robots is intricately connected, characterized by a profound integration of intelligent operations [8]. Moreover, the embodied entities are designed to work alongside human workers, complementing their skills and capabilities. The collaboration between humans and embodied entities plays a critical role in shaping the future of MI. Recent research has equipped these autonomous agents with human-like cognitive mechanisms, such as reflection [18], [19], task decomposition [20], and tool utilization [21]. This symbiotic relationship enhances both human and machine performance, not only by automating repetitive tasks but also by augmenting human creativity and flexibility [22]. The integration of embodied AI systems into various aspects of Industry 5.0 promises to revolutionize the way we live and work, fostering an efficient, adaptable, and innovative future.

Comprised by several agents, the embodied AI systems can bridge the gap between cyberspace and the physical world in Industry 5.0, such that the simulated and physical environments can be interacted in a more human-like manner [23]. With the aid of embodied intelligence (EI) technology, the notions of CPSSs have been shifted to the human-CPSSs. For example, robots can handle dangerous or physically demanding tasks, while humans focus on higher-level decision-making and creative problem-solving. State-of-the-art methods, including visual active perception, embodied interaction, embodied agents and sim-to-real robotic control, have been extensively investigated to address abstract problems in cyberspace and navigate the complexity and unpredictability of the physical world [24]. The multi-modal large models (MLMs) have injected strong perception, interaction and planning capabilities to embodied agents, underscoring their importance in facilitating active interactions in dynamic virtual and physical environments [25]. In MI, the main principles of MLMs for EI lie in three main folds: 1) the fusion of diverse sensory data from different sources; 2) the task decomposition from human instructions and commands; and

TABLE I
THE DEVELOPMENT PROCESS OF INDUSTRIAL REVOLUTIONS FROM INDUSTRY 1.0 TO INDUSTRY 5.0 [8], [14], [15]

Version	Time	Characteristic	Value	Description
1.0	Since 1760s	Age of mechanization	Mechanization, labor division	Machine-based manufacturing
2.0	Since 1870s	Age of electrification	Energy-saving, large-scale	Electric-driven large-scale manufacturing system
3.0	Since 1970s	Age of information	Digital, automative	Information-driven manufacturing system
4.0	Since 2011	Age of industrial internet	Intelligent, interactive	Cyber-physical manufacturing system
5.0	Since 2017	Age of intelligence	Human-centerity, sustainability, resilience	Human-cyber-physical-social manufacturing system

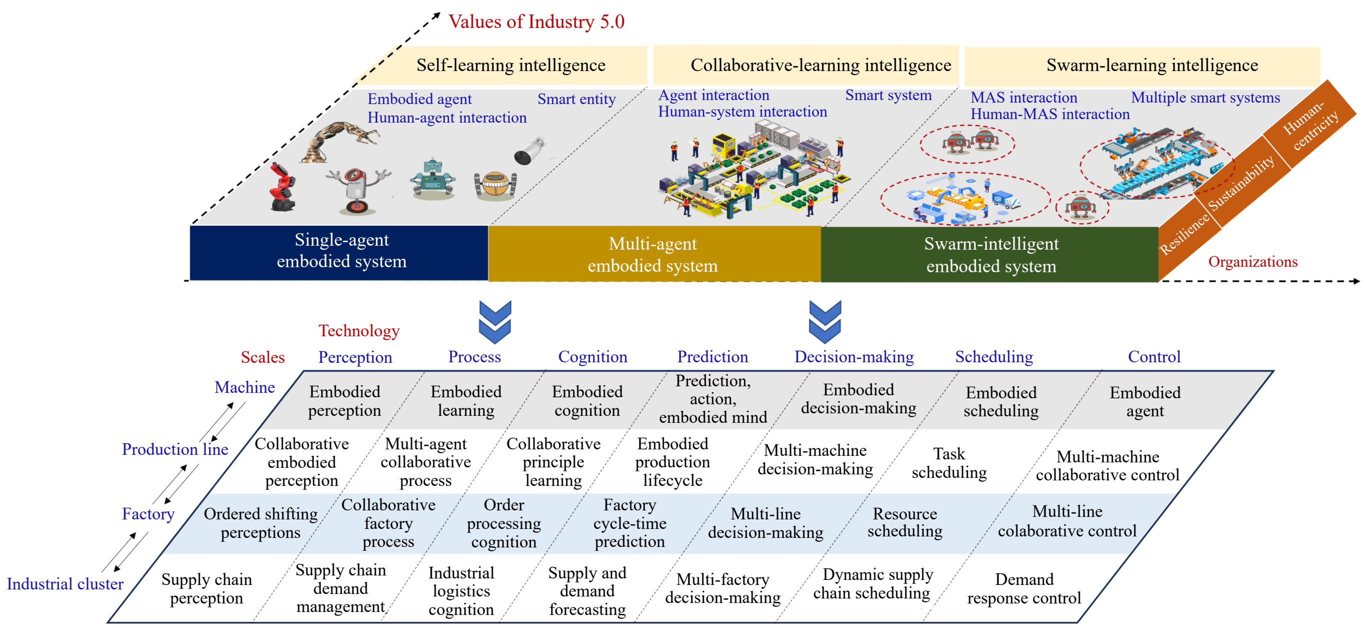

Fig. 2. The framework of smart, lean and agile manufacturing in the context of Industry 5.0.

3) the perspective switching and coordination unification. Recent representative embodied models are RT-2 [16] and RT-H [26], which are designed to learn from large datasets of robot interactions, and generalize this learning to perform new tasks in unseen environments without additional training.

In summary, the main contributions of this work are three-fold. First, a comprehensive framework is presented for smart MI based on the principles of Industry 5.0. To the best of the authors' knowledge, this is the first detailed survey of smart IndAI from the perspective of aligning Industry 5.0 with embodied AI technology, which offers a broad overview, along with a thorough summary and categorization of existing publications. Second, an in-depth investigation of the latest advancements in embodied AI is conducted, which includes comprehensive benchmarking and discussion of current work across multiple simulators and datasets. Our analysis highlights the state-of-the-art techniques and identifies the key areas of improvement. Third, several research challenges and potential directions are proposed for future research in AGI for embodied IndAI.

II. ARCHITECTURE OF SMART MANUFACTURING FOR INDUSTRY 5.0

A. *Different Levels of IndAI*

Industry 5.0 envisions a future of industrial production centering at human-centric and knowledge-driven intelligence for creating value beyond economic benefits. Table I summarizes the major development processes of industrial revolutions from Industry 1.0 to 5.0. Such focus has shifted from mechanic manufacturing, electric manufacturing, digital manufacturing, intelligent and interactive manufacturing to the human-centered, sustainable and resilient manufacturing. Industry 1.0 was characterized by the mechanization of production through water and steam power. Industry 2.0 introduced mass production with the help of electric power. Industry 3.0 saw the advent of electronics and information technology, enabling further automation of production processes. Industry 4.0 brought about the integration of cyber-physical systems, IoTs, and big data into manufacturing, emphasizing smart factories. Different from Industry 1.0 to 4.0, Industry 5.0 introduces a human-centric approach that aims to integrate human skills and creativity with the efficiency and capabilities of automated systems. It focuses on the collaboration between humans and robots or intelligent machines, leading to collaborative robots. The goal is not just to increase productivity but also to enhance the quality of products and services while addressing societal challenges such as sustainability and personalized needs.

Fig. 2 reveals the framework of IndAI grounded in the principles of Industry 5.0, which is designed to foster a human-CPSS MI. There is a strong emphasis on sustainability and

TABLE II
DIFFERENT LEVELS OF INDAI IN INDUSTRY 5.0 [27], [28]

Concept	Size	Organization	Description	Feature
Self-learning IndAI	Small, Single-domain	Smart entity	Robotics, devices Assembly lines, workshops	Embodied tasks Human-machine interaction
Collaborative-learning IndAI	Middle, Single-domain	Smart MAS (Connection of agents)	Assembly lines, Workshops	Multi-agent embodied tasks, Human-MAS interaction
Swarm-learning IndAI	Large, Multi-domain	System of smart MASs (Connection of MASs)	Factories, industrial group, Government, society	Multi-group embodied tasks, Human-group convergent swarm computing

social responsibility, ensuring environmentally friendly production and safe working environments. Differing from general AI, IndAI applies advanced technologies specifically to optimize industrial process [1]. The integration of IndAI with human intelligence creates a powerful symbiosis that enhances productivity, flexibility, and innovation. By adopting augmented decision-making approaches, IndAI can take over monotonous or hazardous tasks, mimicking human intuition for better strategic planning. On this basis, collaborative robots exemplify this partnership, working alongside humans. By using continuous learning, factories can self-optimize and respond rapidly to changes, while preserving valuable human knowledge through systematic capture and transfer. Transparent IndAI further empowers workers by offering insights into operations, fostering trust and acceptance, which drives the development of smarter, safer, and more efficient manufacturing environments, well-prepared to meet contemporary market demands. In Table II, this framework can be structured along three primary dimensions:

1) Single-Agent Embodied IndAI: Self-learning intelligence is required in single-agent embodied IndAI, which refers to the capability of IndAI systems to improve from experience without being explicitly programmed. Embodiment at the single entity level refers to the integration of sensing, control, and perception capabilities to enable the intelligence of embodied entities. This involves the deployment of embodied agents that can sense their environment, make decisions, and execute controls autonomously. At this level, human operators and robotic systems collaborate closely, with robots capable of learning from and mimicking the actions of human workers to optimize production schedules and enhance overall efficiency.

2) Multi-Agent Embodied IndAI: The principles of multi-agent embodied IndAI have been applied in industrial settings, which refers to an intelligent system composed of multiple individual embodied intelligences interconnected through a network topology. At its core, this system is an interconnected intelligence supported and enabled by an information network. Collective embodiment, essentially the networked interconnected intelligence, should be achieved in multi-agent embodied IndAI, which involves the integration of multiple embodied entities (iRobots, devices, workshops, and factories) to form a cohesive multi-agent-system (MAS).

3) Swarm-Agent Embodied IndAI: Swarm embodiment, powered by knowledge and data, drives the interconnected intelligent systems both within and across enterprises. These systems feature self-perception of operating conditions and self-learning of processes, autonomous execution by equipment, and self-organization of the overall system. This level of embodiment aims to optimize the efficiency and environmental sustainability of the entire business operation, from procurement of raw materials to operational decisions, planning, selection of operational parameters, and the execution of production processes. Via industrial internet, various devices share their EI to collaboratively complete production lines or a series of tasks, embodying the collective wisdom of the group.

With three levels of IndAI, Industry 5.0 is integrated with human-centric and technology-driven approaches for creating more sustainable manufacturing processes and fostering long-term economic growth. It promotes a circular economy by facilitating the tracking and reuse of materials, supporting closed-loop systems that reduce dependency on virgin resources. Customization capabilities allow for efficient small-batch production, cutting down on overproduction and waste. Emphasizing knowledge-driven innovation fosters the development of eco-friendly technologies and cleaner production methods. By prioritizing human well-being and continuous skills development, Industry 5.0 enhances worker health and adaptability, supporting a sustainable workforce. Additionally, it enables real-time environmental monitoring and transparent reporting, ensuring compliance with regulations and driving improvements in sustainability practices.

B. Case Study: Embodied Intelligence in Automotive Assembly Manufacturing

Using automotive assembly manufacturing as an example, this section provides a case study that illustrates the progressive application of embodied intelligence from single-agent embodied IndAI to multi-agent embodied IndAI, and eventually to swarm-intelligent embodied IndAI. It demonstrates how EI evolves from executing general tasks and long-term operations to achieving complex factory-wide collaboration. This evolution not only addresses critical challenges such as reasoning speed, resource allocation, and data privacy, but also highlights the hierarchical application value of embodied intelligence in industrial environments.

Single-agent embodied IndAI focuses on specific industrial operations, such as general-purpose units handling tasks with a single robotic arm, inspection devices, and conveyance equipment. For instance, robotic arms are utilized for tasks like body frame welding, painting, parts handling, and assembly. Measurement devices are employed to gauge body dimensions and monitor the positioning of installed components, and automated guided vehicles (AGVs) are responsible for the efficient transportation and distribution of components. In the multi-agent embodied IndAI system, multiple agents collaborate to form dynamic production lines, efficiently completing complex tasks. For example, during the critical wheel installation phase of vehicle assembly, robotic arm A handles wheel grasping and positioning, robotic arm B secures and tightens bolts, while inspection device C monitors bolt torque in real time to ensure secure installation. Multi-agent collaboration not only reduces operational time but also significantly enhances assembly precision.

Extending to the swarm-intelligent embodied IndAI system, automotive assembly manufacturing leverages cloud-edge-terminal collaboration to create a highly interconnected intelligent system. Edge devices, such as sensors on robotic arms, detect subtle changes during the assembly process in real time and upload the data to the cloud for analysis, dynamically adjusting production strategies. The swarm allocation mechanism ensures flexible resource deployment. For example, during peak demand periods, AGVs and robotic arms can be reallocated from lower-priority production lines to critical ones to alleviate production pressures. Moreover, computation offloading techniques enable complex tasks, such as predictive maintenance algorithms, to be distributed between edge devices and the cloud for collaborative processing, further improving production efficiency. Through federated learning, sensitive production data can remain local, enabling collaborative learning among distributed devices while protecting data privacy and optimizing global production workflows.

This multi-level framework supports multi-line and multi-factory coordination, enhancing not only assembly efficiency but also optimizing other industrial processes, such as material scheduling and supply-demand matching.

III. Single-Agent Embodied IndAI

There has been an emerging paradigm shift from the era of “data-based IndAI” to “embodied IndAI”, where AI algorithms and agents are transitioning from learning primarily from datasets of images, videos, and text to learning through direct and egocentric interactions with environments in a human-like fashion [23]. As the crucial carriers of embodied IndAI, an iRobot is an autonomous entity that observes its environment through sensors and acts upon it using actuators, which not only perform traditional manufacturing tasks such as chip assembly, product inspection, and material handling, but also achieve self-optimization and personalized services [29]. For example, the practical use of intelligent robot manipulators can improve the efficiency and performances of robotic assembly lines. Fig. 3 presents a comprehensive overview of how embodied AI can be applied in robotics, showcasing a variety of robot types such as fixed-base, wheeled, quadruped, tracked, humanoid, and biomimetic robots [30]. The key components of embodied robotics are classified into three main areas: 1) embodied perception, including technologies like visual simultaneous localization and mapping (SLAM), semantic segmentation [31], and depth estimation [32]; 2) embodied execution, involving path planning [33], obstacle avoidance, and goal-directed movement [34]; 3) embodied scheduling, focusing on multimodal data fusion, vision-language models (VLMs), and task execution. Several application challenges have been put forward, such as handling general-purpose and long-horizon tasks, improving inference speed, and operating in high-dynamic environments.

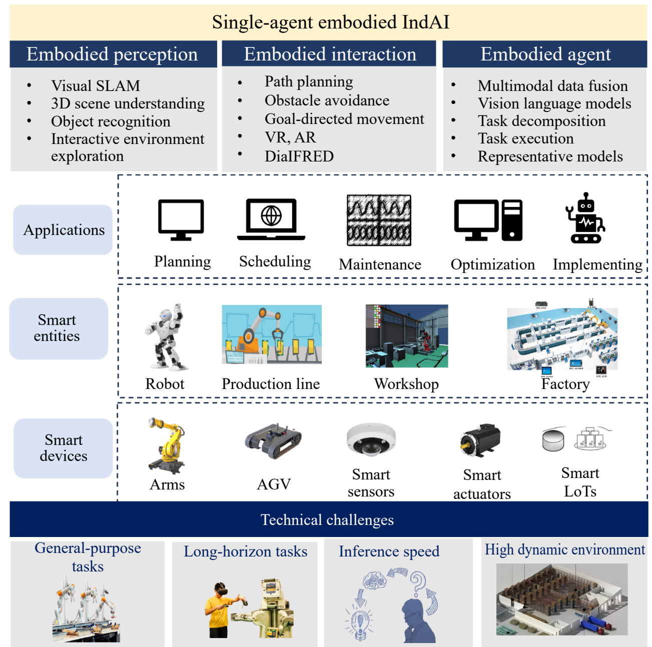

Fig. 3. The key aspects of single-agent embodied IndAI in Industry 5.0.

The key intelligent technologies of iRobot as a single agent includes environmental sensing [35], data processing, robot control, AI-driven decision-making, and human-like cognition. The rapid development of large language models (LLMs) [36], [37] and VLMs [38], [39] has significantly advanced industrial production. The most cutting-edge approaches now leverage VLMs capable of processing multimodal inputs, such as text, images, or video, to integrate human instructions/text data and visual data. VLMs, such as OpenAI’s GPT-4(V) [40], [41], have made significant strides in understanding and processing both textual and visual information. Such models are increasingly being applied in the field of industrial robotics, enhancing the robots’ cognitive and operational capabilities for handling complex tasks in dynamic environments. This, in turn, improves the precision and adaptability of industrial robots in tasks like grasping, moving, and positioning.

To achieve human-robot collaboration, single IndAI must address two critical points: whether the agent possesses generality, and its ability to execute long-duration tasks. However, in practical applications, achieving generalization and executing long-term tasks often proves challenging. Regarding generalization, learning-based robotic operations typically rely on training within simulators, which are constrained by limited categorical data, making effective generalization in diverse object categories and complex scenarios difficult. At this stage, it is crucial to explore how to leverage VLMs to enhance the generalization capabilities of industrial robots. This enhancement enables them to adaptively execute complex tasks in dynamic environments and seamlessly collaborate with human workers, aligning with the goals of Industry

5.0 to create more efficient and personalized production processes.

A. General-Purpose Tasks

In this section, the issue of generalization in robotic operations within industrial production is addressed. Industry 5.0 demands greater flexibility and production efficiency within complex workflows to meet diverse and personalized requirements. The technology of VLMs integrating human knowledge, zero-shot learning, and teaching-based learning with diffusion policies, can significantly enhance the model's generalization performance. On this basis, robots can participate more intelligently in production processes, improving precision and efficiency in tasks such as grasping, moving, and positioning, thereby laying the foundation for future smart manufacturing and automated production.

In [42], [43], how to imitate human behavior by selecting optimal positions for object grasping was discussed. CoPa [42] utilized common-sense knowledge from foundational models to generate sequences of 6-DoF end-effector poses for open-world robotic manipulation. Some methods generated actions directly from images [27], [44]–[46]. Certain works achieved better control through fine-tuning [27], [44]. In [44], RT-2 was based on fine-tuning large VLMs trained on web-scale data for generalized and semantically aware robotic policies. This involved adapting VLMs, initially designed for tasks like open-vocabulary visual question answering and visual dialogue, to output low-level robotic actions directly. ManipLLM [27] fine-tuned multi-modal LLMs (MLLMs) to predict 3D contact points based on text prompts, RGB images, and depth map inputs. The gripper's upward and forward directions were predicted from the SO(3) rotation for the end-effector. Some methods did not require fine-tuning [45], [46]. MOKA [45] introduced a point-based affordance representation that linked VLM predictions from RGB images to the robot's physical motions. OK-Robot [46], by integrating VLMs for object detection, navigation primitives for movement, and grasping primitives for manipulation, provided a comprehensive pick-and-drop solution without additional training.

In industrial environments, occlusions and complex object arrangements are common, making robotic grasping in cluttered settings a significant challenge. Even the existing methods often struggle to accurately identify and grasp target objects that are severely occluded or completely hidden. However, ThinkGrasp in [47] addressed this issue by effectively identifying and generating grasp poses for target objects, even when they were heavily obstructed or nearly invisible, using goal-oriented language to guide the removal of obstructing objects. Policy learning from demonstration is crucial in industrial processes where the agent receives user prompts and an image of the current state. The skill selection module, implemented through a foundational model, selects an appropriate skill to execute the task. If no suitable skill is available, the agent requests the worker to perform several demonstrations and utilizes visuomotor diffusion policies to train a new skill. This approach allows the agent to continuously learn and adapt to diverse tasks, improving its efficiency and flexibility in handling complex industrial scenarios. Diffusion policy [48] introduced a novel approach to generating robot behavior by representing a robot's visuomotor policy as a conditional denoising diffusion process. In [49], 3D-VLA was built upon a 3D-based LLM to incorporate a set of interaction tokens with the embodied environment. To enhance the model's generative capabilities, we integrate a series of embodied diffusion models with the LLM, enabling the prediction of goal images and point clouds.

B. Long-Horizon Tasks

For long-duration task execution, long-sequence task planning faces numerous challenges, including temporal dependencies and combinatorial complexity, as current actions significantly influence future decisions and outcomes. Additionally, the forgetting problem, increasing the difficulty of maintaining decision coherence, may make agents lose effective control over previously acquired information. Environmental changes necessitate that agents dynamically adjust their objectives and strategies, while the increase in computational complexity places higher demands on a large amount of resources and computational powers. Therefore, agents need to possess both stability and flexibility to effectively address complex long-term tasks in the context of Industry 5.0. Complex long-horizon planning tasks require extensive context, including instructions and demonstrations, to ensure that the generated plan can be executed accurately. However, LLMs may sometimes overlook or misinterpret the rules provided, leading to invalid plans or even dangerous actions. Moreover, such types of tasks are highly prone to error propagation, where mistakes in early stages may easily cause deviations in subsequent steps, further compounding the problem. Successfully addressing complex interactive tasks requires agents to demonstrate long-horizon planning, long-term memory retention, subgoal decomposition, spatial reasoning, exception handling, and commonsense knowledge. FLTRNN [50] generated a task plan from a given goal by breaking down complex tasks, solving subtasks with language-based RNNs and memory, and enhancing reasoning through Rule Chain-of-Thought and a memory graph. The dual-process theory suggested that human cognition involved both fast, intuitive thinking and deliberate, and analytical reasoning. Fast thinking is similar to seq2seq methods, which rely on imitation and shallow patterns, while deliberate reasoning resembles LLMs. Swiftsage [51] combined these approaches, integrating the strengths of behavior cloning and LLM prompting to improve performance and efficiency in complex tasks.

In industrial automation, especially in tasks such as robotic manipulation, production line control, and quality inspection, the ability to perceive and reason about physical states is essential, a capability that LLMs currently lack. However, with the development of VLMs, these models can now seamlessly integrate perceptual data into reasoning and planning processes, enhancing robots' understanding of commonsense knowledge in industrial environments. This enables robots to make more informed decisions and perform operations more effectively in complex scenarios, such as recognizing object states, positions, and functions, which in turn optimizes production efficiency, improves product quality, and reduces reliance on human intervention. The advancement of VLMs offers a promising solution for addressing complex perception challenges in industrial settings, where visual comprehension must be tightly integrated with operational decision-making. RoboEXP [52] focuses on robot exploration and active perception, emphasizing the ability to explore environments, particularly to locate items that are not immediately visible. This enables robots to autonomously explore environments and generate action-conditioned scene graphs (ACSGs). Zhu *et al.* [53] improve on existing methods by using a Think Bank to process tasks through fast or slow thinking, with simple tasks handled by a policy network and complex tasks managed by a VLM for reasoning and execution.

C. Inference Speed

In industrial manufacturing, improving the response speed of models is crucial. However, the inference speed of current models is relatively slow, primarily due to the following factors: 1) Large model size: Larger models demand substantial memory, memory access, and computational resources. For instance, the FP16 weights of a 175B GPT-3 model require 350 GB of memory, which necessitates at least five 80 GB A100 GPUs just to store the model. Even with sufficient GPUs, the extensive memory access and computation significantly slow down inference. 2) Attention operation: In the prevailing transformer architecture, the attention mechanism is I/O-bound, with a quadratic memory and computational complexity concerning sequence length. 3) Sequential decoding: During inference, models generate only one token at a time, which introduces significant latency, because this process cannot be parallelized. Reference [54] extensive researches have been conducted to address the first and second challenges for optimizing large model size [55]–[58] and improving attention operations [59]–[61] — through methods like model compression/redesign [55], [58]–[61], serving system optimizations [56], and hardware advancements [57]. Ning *et al.* [54] tackle the third issue by questioning the necessity of fully sequential decoding. They propose the feasibility of parallel decoding without altering existing models, systems, or hardware. Their idea stems from how humans answer questions. Humans do not always think and write answers sequentially. Instead, for many types of questions, we first outline a framework based on certain protocols and strategies, then add evidence and details to explain each key point.

Recently, approaches focusing on sparse and robust structural representation learning have shown promise in addressing inference speed issues, such as [62], [63]. These methods aimed to reduce redundant computations by leveraging structural properties of graph-based data. These methods reduced redundant computations by leveraging the structural sparsity of graph-based data, and improving the efficiency of IndAI systems. By minimizing memory access and computational overhead, they addressed critical bottlenecks in applications like real-time monitoring and predictive maintenance. Additionally, optimized attention mechanisms enhanced the processing of large-scale, dynamic graphs, enabling faster inference and making IndAI models more suitable for time-sensitive industrial applications.

IV. Multi-Agent Embodied IndAI

Multi-agent embodied IndAI refers to the notion of "population thinking", where the adaptivity of entire populations is studied rather than that of individuals. A MAS, essential for AGI, is composed of multiple interacting intelligent agents. This integration bridges the gap between cyberspace and the physical world, enabling more sophisticated and adaptive systems. The emergence of IoT and the deployment of computing systems have led to the formation of embodied MASs to complete a variety of tasks. IoT devices, such as sensors and actuators, collect real-time data from the environment. IoT devices can monitor various parameters, including temperature, humidity, light, and motion. The data collected is crucial for the agents to understand their surroundings and make informed decisions. Then, computing systems integrate these physical sensors with computational resources, allowing for real-time processing and analysis of the data, ensuring that the agents can react swiftly to changes in dynamic environment.

The most important element, the communication network and IoT, is required for implementing the decentralized and distributed algorithms of robust sensing and perception, formation tracking, task allocation, path planning and navigation, and human-robot interaction. The implementation of multi-agent embodiment in EMRSs should be used under the communication constraints, the topology change, the uncertain environment. Based on smart IoT and communication network, the agents in a EMRS can also be physical robots, humans, robot teams, or even human teams. A notable feature of EMRSs is the capability to form combined human-agent teams, leveraging the strengths of both human and artificial entities. Specifically, the complex tasks can be decomposed and allocated to specialized iRobots, each handling sub-tasks that best match their capabilities. Enhanced perception and information gathering are achieved through sensor fusion and distributed sensing, providing a more comprehensive and accurate understanding of the environment. Additionally, these systems are finely cooperated and interacted by using the decentralized and distributed control [64]–[66], with the aim of enhancing safety and security by mitigating risks in hazardous environments, and provide better monitoring and threat response. Finally, the multi-robot collaboration improves decision-making and planning through distributed and flexible approaches.

In Fig. 4, an overview of embodied MASs is presented, focusing on the interaction and coordination among embodied agents in an industrial setting. Different from single-agent embodied IndAI, the embodied MAS is the integrated system composed of several embodied agents. Since the traditional centralized training may be unsuitable for data-driven industrial scenarios, large centralized systems have seldomly been used in multi-agent embodied MI systems [2]. Instead, the distributed AI is significantly required for treating each agent as an artificial entity that is capable of perceiving the surroundings using sensors, making decisions, and taking actions by using actuators [67]. The combination of humans, robots, and products should be considered for the embodied multi-iRobot

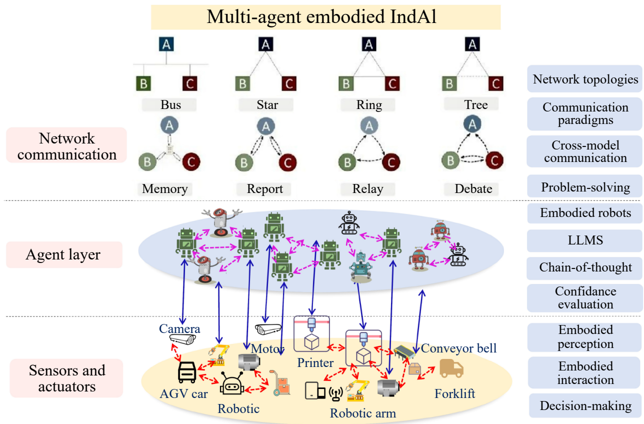

Fig. 4. The framework of multi-agent embodied IndAI in Industry 5.0.

systems (EMRSs) [68], [69], product status sensing systems, and process control systems. Three main elements of EMRSs, the type of agents, control architectures, and communications, were discussed in [2].

A. Autonomous Multi-Agent Learning

The internet serves as the backbone for communication between different agents and the central control systems. High-speed, reliable internet connections enable seamless data exchange, allowing agents to coordinate their actions and share information. Multi-agent communication protocols, such as message queuing telemetry transport (MQTT) and constrained application protocol (CoAP) [70], ensure that agents can communicate efficiently and securely.

On the other hand, distributed computing environments are becoming the basis for the multi-robot collaboration. In this case, the communication network, however, maybe disconnected at times and its links may have varying strengths. The network bandwidth is often considered a bottleneck for distributed AI. Once the bandwidth is fully occupied, some agents may be unable to send messages promptly to others, leading to decision delays and impairing cooperative effectiveness. In an unstable communication network, the teamwork in multi-iRobot systems was enhanced by using embodied intelligence via model-and data-driven approaches [71]. The active perception in situated multi-iRobot systems under topology switching was addressed in [72]. Game-theoretic multi-iRobot control and network cost allocation was studied in [73] under unstable data transmission. In the dynamic obstacle environment, the collision-avoidance path planning mechanism of multi-robot systems was designed in [74]. The distributed communication-aware task allocation was investigated in [75] for EMRSs from STL and SpaTel specifications.

To further enable a highly intelligent team behavior, the end-to-end multi-robot collaboration involves training robots to work together autonomously, from perception and decision-making to execution, without the need for explicit programming for each step. The multi-agent reinforcement learning (MARL) for distributed multi-iRobot collaboration is ever growing. The multi-agent versions of all algorithms (i.e., MASAC, MADDPG, MAPPO) were introduced in [76]. Event-triggered communication network with the limited-bandwidth constraint was designed in [71] for MARL. In [77], using behaviors as the underlying control representation provided a useful encoding that enhances robustness in control and allows abstraction, making it easier to handle scaling in learning, particularly in EMRSs. An information-theoretic and fully decentralized cooperative MARL framework, called InfoPG was discussed in [78], where agents refine their action decisions step-by-step based on the actions of their teammates.

B. Heterogeneous Multi-Agent Learning

Heterogeneous multi-robot collaboration using IoT technology represents a significant advancement in the field of robotics and automation. In such collaborations, robots with different capabilities, sizes, and functionalities are interconnected through an IoT network, allowing them to share data, coordinate tasks, and operate in a coordinated manner. For example, in a smart factory setting, a combination of ground robots for material transport, aerial drones for inspection, and stationary robotic arms for assembly can all be integrated into a single system. IoT sensors and communication protocols facilitate real-time data exchange, enabling robots to adapt to dynamic changes in the environment and adjust their actions accordingly. By leveraging the strengths of heterogeneous robots and the connectivity provided by IoT, these systems can tackle a wide range of challenges in manufacturing, industries, leading to more resilient and adaptive automation solutions.

The challenges of heterogeneous MAS sub-fields including task decomposition, coalition formation, task allocation, perception, and multi-agent planning and control was summarized in [79]. A significant part of researches on multi-agent IndAI concerns reinforcement learning techniques. Multi-agent deep reinforcement learning is always used in heterogeneous MASs, where multiple agents not only learn from their own experiences, but also from each other in EMRSs. The heterogeneous policy (HetNet) was designed to learn efficient and diverse communication models for coordinating cooperative heterogeneous EMRSs. The distributed framework was proposed in [80], which enabled a team of heterogeneous robots to rapidly generate actions from a common and user-defined goal specification. The framework for resilience in a networked heterogeneous EMRS was proposed in [81], in presence of resource failures and corruptions.

Recent advancements in LLMs have significantly enhanced their capabilities in complex reasoning tasks, particularly in the context of heterogeneous multi-robot collaboration. One notable development is the introduction of the chain-of-thought (CoT) technique, which has improved the ability of LLMs to break down and solve intricate problems. Building on this, a novel framework called exchange-of-thought (EoT) was proposed in [82] to facilitate cross-model communication during problem-solving. EoT employed four communication paradigms: Memory, report, relay, and debate. These paradigms enable different models to share insights, coordinate actions, and refine solutions collaboratively. Additionally, a series of self-correction methods have been developed in [83], aiming to iteratively improve the quality of answers by leveraging the models' feedback on previous outputs. These techniques collectively enhance the robustness and effectiveness of EMRSs, making them more adaptable and efficient in dynamic and complex environments.

C. Adaptive and Flexible Multi-Agent Learning

Adaptive and flexible multi-robot collaboration is a critical area of research that focuses on enhancing the coordination and interaction among diverse robotic systems in dynamic and unpredictable environments [84]. One of the key components is dynamic task allocation, where robots can autonomously assign and reassign tasks based on real-time conditions and resource availability. This ensures that the most suitable robot is always handling the most appropriate task, optimizing efficiency and effectiveness. Real-time communication is another essential aspect, often facilitated by the IoT, allow robots to share data and coordinate actions in real-time. This capability is crucial for adapting to sudden changes in the environment or unexpected events. Advanced sensors and perception systems provide robots with situational awareness, enabling them to make informed decisions and react appropriately to new information. This situational awareness is vital for navigating complex and dynamic environments.

On the other hand, human feedback help to refine the performance of EMRSs, making the overall system more adaptive and flexible. Human-in-the-loop EMRSs are hybrid strategies to incorporate humans' teaming strategies for robot teams and directly learn multi-iRobot collaboration policies from human experts through developing methods for multi-agent learning from demonstration (MA-LfD). Effective collaboration requires humans and robots to exchange information through the communication network to achieve shared objectives [85]. Recent progress in LLMs [37], [86]–[88], particularly GPT-4 [37], has opened new pathways in this domain. These LLMs are highly proficient in understanding human intent, executing commands, and driving the development of Industry 5.0. Their demonstrated capabilities span across language comprehension, vision, and coding [89]. By leveraging LLMs, autonomous agents are empowered to make nuanced decisions and carry out tasks with unprecedented autonomy.

In [90], the collaborative site inspection and target classification was designed between the operator and the EMRS, under the limited query opportunities and poor communication channel quality. The human-robot communication for collaborative decision-making was extensively discussed in [91]. In [92], a cooperative tracking control for heterogeneous Euler-Lagrange systems in human-robot cohabitation was developed, accounting for communication delays and relative positioning constraints. The fatigue model of humans, introduced in [93], was applied to a mixed team coordination framework to predict team performance under the constraints of human fatigue. To further enhance Human-EMRS collaboration, [94] proposed gradient compression to train LLMs using common computing resources with low bandwidth and unstable networks, thereby optimizing the training process and reducing computational costs.

V. SWARM-INTELLIGENT EMBODIED INDAI

Single agents or small-scale clusters are insufficient for the demands of Industry 5.0. Similar to human collectives, robots must collaborate as a group to realize their full potential. Industrial swarm intelligence (SI) refers to the application of SI technology in industrial environments to achieve more efficient, flexible, and intelligent production processes. SI is a method that studies how to mimic the collective behavior of simple organisms in nature, such as ants, bees, and birds, to solve complex problems. Although these organisms individually exhibit low levels of intelligence, as a whole, they can display highly complex intelligent behaviors, such as foraging, nest building, and migration. The blue picture of SI-based smart factory has been shown in Fig. 5. The principles of SI can be applied to dynamic multiobjective optimization problems to enhance search efficiency, robustness, and adaptability. The method of non-inductive transfer learning was discussed in [95] to exploit knowledge from previously encountered optimization landscapes without assuming direct correlation. In [96], the adaptive mechanism algorithm was incorporated with multiple strategies for exploration and exploitation, adapting its behavior by the current state of the optimization process.

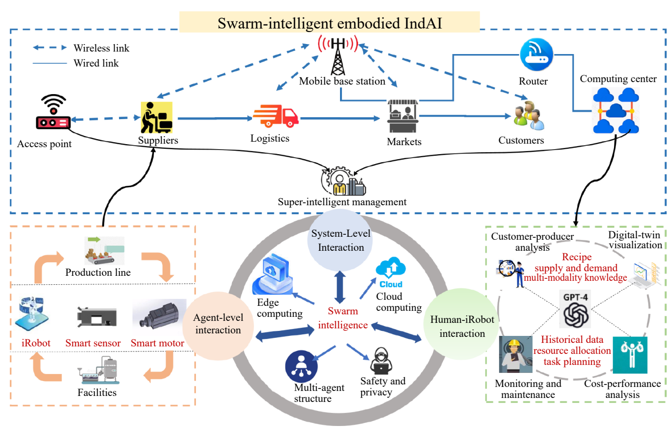

Fig. 5. The concept of swarm intelligence in Industry 5.0.

Multi-agent embodied IndAI refers to a system where intelligence emerges from the interactions of a relatively small number of embodied agents [97]. The MAS can operate with either centralized or decentralized control mechanisms under a small network architecture [98]. Differently, SI embodied IndAI refers to the system of embodied MASs, such that the intelligence emerges from the collective behavior of MASs. SI systems are always built with cloud-edge-terminal (CET) structure for organizing a group of MASs. As progresses toward AGI and Industry 5.0, extensive researches are being conducted to explore the concept of assembling these agents into groups or societies, examining the potential benefits of their collaboration. In the framework of SI embodied IndAI, the CET technology should be used to enhance SI in dynamic environments, and optimize applications such as computing offloading, resource allocation, and security management. Thus, the CET system in an intelligent factory has multiple edge servers and terminal devices. In this section, we will explore two types of EI groups, divided into LLM-based role distribution and multi-body systems based on machine entities.

A. Cloud-Edge-Terminal Collaborative Training

By leveraging the autonomy, coordination, and adaptability of agent swarms, Industry 5.0 can be achieved with greater flexibility, efficiency, and sustainability. This collaborative CET model optimizes data processing, reduces latency, and improves reliability and flexibility, making it ideal for Industry 5.0 applications that require real-time responsiveness and large-scale data analysis. The terminal layer consists of sensors, smart devices, and user interfaces, responsible for collecting real-time data such as temperature and machine status. The edge layer processes data received from the terminal layer. Devices, such as edge servers and gateways, and located near terminal devices, handle time-sensitive tasks such as equipment monitoring and real-time fault detection, minimizing reliance on cloud processing. The cloud layer primarily consists of cloud computing architecture, including computing units, storage units, and data centers. Although the edge layer can perform computations, tasks requiring more intensive processing must still be uploaded to the cloud layer for further unified and intensive handling.

The cloud model, with its strong computational capabilities, assists edge models in prediction and performs slower but comprehensive reasoning. In return, edge devices rapidly sense and make accurate identifications, transmitting real-time feedback to the cloud. A key research focus is how cloud, edge, and terminal systems exchange prior or privileged information to collaboratively meet environmental demands. There are several challenges with cloud-edge collaboration in practical applications. First, the decision-making response time depends on computation and communication latency. Second, lifelong learning and adaptability are crucial, as embodied agents must continuously interact with dynamic environments, requiring the ability to self-learn and adapt based on perception and feedback. Therefore, achieving real-time, fast response, dynamic, scalable, and secure cloud-edge collaboration is vital.

To address slow response times and enhance scalability and practicality, lightweight models can be deployed on embodied agents. Common model compression techniques include model pruning [99], [100], model quantization [58], [101], [102], and knowledge distillation [103], [104]. SparseGPT [99] tackled pruning for GPT models by framing it as a large-scale sparse regression problem, eliminating the need for retraining. LLM-Pruner [100] selectively removes non-essential structures based on gradient information, introducing the first structured pruning method for large models. SmoothQuant [58] addressed outlier activations in large language model quantization by smoothing the channel amplitudes through channel-scale transformations, making quantization easier. LLM-QAT [101] improved throughput by quantizing model weights, activations, and the key-value cache. SSD-KD [103] models and transfers relationships both between neighboring sample pairs and across different channels of each sample for better knowledge transfer. Fine-tune-CoT [104] enhanced reasoning capabilities in smaller models by fine-tuning with reasoning samples generated by large models, demonstrating substantial improvements in inference performance.

For the challenges posed by new domains and variations in data distribution at the edge, improving the generalization capability is essential for handling new and diverse environmental conditions. Ma *et al.* [105] explored hierarchical memory replay with continual learning, enabling dynamic allocation of training resources for continual learning. Nan *et al.* [106] proposed a cloud-edge collaborative learning architecture that integrates edge computing with cloud-based continual learning. Lightweight models are deployed at the edge, while large cloud models continuously train them to improve efficient video analysis capabilities, reducing data transmission between the edge and the cloud. To facilitate knowledge transfer between models, Daga *et al.* [107] introduced the CLUE learning framework, which dynamically extracts important parameters from auxiliary neural networks and uses multi-model enhancement to improve the predictive performance of the target nodes.

B. Dynamic Task-Agnostic Swarm Allocation

In human-centered industrial production processes, there is often a heavy reliance on human intuition and consultation. Recent advancements in deep learning have introduced precision designs across various stages of industrial production, significantly transforming the industry. With the emergence of LLMs, many applications are beginning to employ multiple LLM agents. In a multi-agent framework, these agents assumed different roles, communicating in natural language to collaboratively tackle specific tasks. An innovative paradigm can be proposed that leverages large models throughout the entire industrial production process, simplifying and unifying communication and production steps, thereby eliminating the need for specialized models for each production phase. In the field of industrial manufacturing, multiple LLM agents take on different roles [108]–[112], communicating in natural language to collaboratively solve a given task, ranging from a simple assembly task to production coordination within a specific area. This process can be systematically divided into four distinct stages: design, production, testing, and optimization. Each stage engages a set of production agents, such as designers, production monitors, and testing engineers, facilitating collaborative dialogue and ensuring smooth workflows. An intelligent scheduling system serves as a coordinator, breaking each phase into atomic subtasks. This approach allows roles to both propose and verify solutions, efficiently addressing specific subtasks through context-aware communication.

In Industry 5.0, multi-agent collaborations require a systematic framework to improve generalizability, efficiency, and performance. Such a framework should include the following features: 1) A task-agnostic system: The system should not rely on task-specific tools but instead be able to generalize across different domains. A task-agnostic system would allow for rapid adaptation to new production scenarios and tasks. 2) Enhanced production efficiency: It should be capable of dynamically removing underperforming agents, which would reduce costs and prevent these agents from generating unnecessary noise that could affect the team's performance. 3) Dynamic team optimization: Effective role distribution is critical for optimal task execution. A well-optimized system should be able to autonomously adjust the roles of agents based on the task requirements, minimizing the need for extensive human supervision.

LLM agents have already demonstrated effectiveness in industrial production processes. By combining multiple LLM agents, their performance can be further enhanced. Existing methods employ a fixed set of agents that interact in a static architecture, limiting their generalizability to different production tasks and requiring each agent to be designed based on specific production tasks, relying heavily on human prior knowledge. Wherein, LLM-agent teams were often configured statically with predefined roles for each agent, such as assembling specific parts or acting as a tester, communicating sequentially in a fixed order to improve code generation [109]. Park *et al.* [112] identified emergent social behaviors in multi-agent life simulations, while the studies of [28], [97], [111], [113], [114] highlighted the decision-making capabilities of MASs on collaborative problem-solving tasks.

However, these studies have focused on specific and narrow tasks, limiting the generalizability of their findings. Additionally, the static nature of agent collaboration — where roles and capabilities remain rigid — hampers adaptability in dynamic environments. Although swarm-agent collaboration improves operational efficiency, using a static setup with predefined roles in industrial production presents several issues: 1) manual setups require designing task-specific roles for LLM agents in each particular domain [109], making it difficult to generalize across different domains or tasks; 2) LLM agents interact in a fixed order to generate answers [18] or engage in single-round interactions [115], which prevents them from achieving optimal solutions; and 3) predefined roles rely heavily on human priors for design and may not align with actual situations [116]. Therefore, some approaches attempt to construct a strategic team of agents that communicate in a dynamic interaction architecture based on task queries. DyLAN [116] is a generic LLM-agent collaboration framework with a dynamic multi-layered feed-forward network, incorporating inference-time agent selection, early-stopping, and automatic team optimization via an unsupervised Agent Importance Score, demonstrating high accuracy, efficiency, and stability in reasoning and code generation tasks. AGENTVERSE [117] general multi-agent framework simulates the problem-solving procedures of human groups, and allows for dynamic adjustment of group members based on current progress. Zhuge *et al.* [118] unifies language agent systems as optimizable computational graphs, introduces an open-source framework for constructing agent systems by recombining fundamental operations, and develop optimization methods for nodes and edges to enable automatic improvements in agent prompts and orchestration.

On the other hand, advanced machine learning techniques for social computing, such as graph learning and embedding, significantly enhance the performance of multi-agent embodied IndAI. In [119], a method that integrated graph clustering as an auxiliary optimization objective into graph attention networks was designed to enable dynamic interaction modeling among agents, thereby improving collaboration efficiency and adaptability. Similarly, the free-rider and conflict-aware collaboration framework for cross-silo federated learning reported in [120] offered strategies to balance cooperative and competitive interactions for fair resource allocation and improved task performance in MASs. By incorporating advanced techniques such as graph attention clustering [119] and collaboration-aware modeling [120], these frameworks produced more effectively model social interactions among agents, unlocking their scalability and potential for broader industrial applications.

C. Computing Offloading and Resource Allocation

In CET, most AI-driven applications require high-performance servers to handle complex computational tasks, which can lead to significant energy consumption in industrial IoT environments [6]. The evolving landscape of artificial IoT presents a series of intricate challenges, particularly in the realm of resource allocation across the cloud, edge, and terminal layers: 1) the centralized processing in cloud computing suffering from higher latency and bandwidth constraints, less suitable for real-time applications, 2) the decentralized processing in edge computing suffering from scalability, limited computational and storage capabilities, 3) the health maintenance of terminal devices with cheap sensors and actuators.

Mobile-edge computing (MEC) has emerged as a prominent solution to deliver mobile services with high computational requirements. By deploying computing resources closer to the end-users at the network's edge, MEC reduces latency, enhances data processing speed, and improves overall service quality, making it particularly suitable for real-time and data-intensive applications. In [121], the integration of nonuniform quantization and asymptotic federated learning (FL) was used to facilitate CET collaboration in artificial IoT, which enabled model training without centralizing data. Federated deep reinforcement learning was proposed in [122] for energy-efficient edge computing offloading and resource allocation in industrial internet. The mixed integer non-linear programming method was proposed in [6] to jointly decide the execution scheduling of the computation tasks (i.e., the device where each task is executed) and the computation/communication resource allocation. Moreover, the MEC of multi-user, multi-server multi-edge computing system, consisting of a number of multiple devices and multiple base stations, should be considered. The mixed integer non-linear programming problem was investigated in [123] to jointly decide the execution scheduling of the computation tasks (i.e., the device where each task is executed) and the computation/communication resource allocation. In [122], the edge caching was addressed in mobile networks to support the large-scale MEC, and the rapidly growing IoT services and applications.

D. Data Security and Privacy

As the transition toward Industrial 5.0 emphasizes a human-centric, secure, and resilient interaction between humans, intelligent systems, and cyber-physical environments, addressing privacy and security concerns is critical in swarm-agent collaboration scenarios [124]. SI systems bring unique risks due to the complexity of interaction environments and the usability of tools. The problem of data security and privacy of LLM-based agents should be investigated from three perspectives: agent quantity, role definition, and attack level, highlighting the need for enhanced trust and transparency in swarm-intelligent systems under the Industrial 5.0 framework.

In large-scale clusters, ensuring security and privacy during collaborative training, inference, and task allocation is of utmost importance. Current large models fall under the category of generative models, making it challenging to verify the authenticity of generated content. This often led to hallucination issues, posing growing safety concerns [125]. In the context of industrial production, it is crucial to explore methods to prevent such problems. For example, SelfCheckGPT [125] tackled the hallucination issue by adopting a simple yet effective approach: if a large language model had a solid understanding of a particular concept, its sampled responses should be consistent and factually accurate.

Moreover, as Industry 5.0 systems increasingly integrate autonomous decision-making and adaptive behaviors, safeguarding sensitive operational data becomes critical. Advanced encryption techniques, such as homomorphic encryption, can offer a solution by enabling secure computations on encrypted data without decryption, thus ensuring data privacy even in shared environments. In CET, information leakage is a significant concern that can be effectively mitigated through FL [126]. FL enables a secure and private distributed learning process by allowing multiple edge devices to collaboratively train a model without sharing raw data, thus preserving user privacy. However, FL is not immune to various types of attacks, including Byzantine behaviors, where malicious participants can introduce erroneous updates to disrupt the training process. To address the issue of location exposure during data uploads by edge-side small models, a secure federated submodel learning scheme was designed to be integrated with a private set union protocol [127]. This approach ensures that the identities and locations of participating devices remain confidential, thereby enhancing the overall security and privacy of the federated learning process. By combining robust security measures with efficient privacy-preserving techniques, a comprehensive solution was proposed to the challenges of information leakage and malicious attacks in edge-cloud collaborative training.

To sum up, the Industry 5.0 vision demands a holistic approach to data security and privacy that spans across all layers of industrial systems. This includes securing data in transit, storage, and processing, as well as implementing trust mechanisms to verify agent actions and interactions. By combining robust security measures with efficient privacy-preserving techniques, comprehensive solutions can address the challenges of information leakage, malicious attacks, and accountability in edge-cloud collaborative training. These efforts ensure that data security and privacy are upheld as foundational principles in the advancement of Industry 5.0.

VI. Future Work

Traditional manufacturing faces challenges such as limited manufacturing tasks, structured manufacturing environments, inefficient human interactions, and a lack of collaboration among intelligent agents. In Fig. 6, the relationship between EI and Industry 5.0 is characterized by human-centric innovation, enhanced enterprise resilience [128], and environmental sustainability. EI, leveraging the principles, knowledge, data, and experience of the industrial world, uses large and small models to break down information silos between processes and fill the gap in knowledge across multiple scenarios. This collaboration allows workers to focus on creative and problem-solving tasks while EI handles repetitive and dangerous jobs, promoting a more human-centered approach. Additionally, EI boosts enterprise resilience by dynamically responding to real-time changes, optimizing production, and mitigating risks, thus ensuring operational continuity in the face of supply chain disruptions and market volatility [129]. Furthermore, EI supports sustainability goals by optimizing resource use, reducing waste, and promoting circular economies through precise and automated processes [130]. By integrating EI, Industry 5.0 not only enhances efficiency but also contributes to a more balanced and sustainable future. Future works will be focused on:

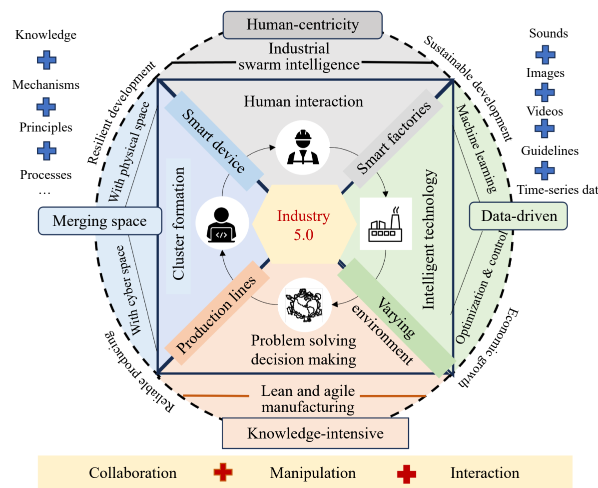

Fig. 6. Relationship between embodied AI and Industry 5.0.

1) Model Layer: In the context of Industry 5.0, the model layer serves as a critical component that bridges the physical and digital worlds, enabling intelligent control and optimization of industrial processes. This layer encompasses various algorithms and technologies that are essential for enhancing system efficiency, reliability, and safety.

a) Model Interpretability and Explainability: Industry 5.0 emphasizes human-machine collaboration, with the IndAI models used to facilitate autonomous interactions in dynamic environments. However, the issue of model hallucination — where systems make erroneous decisions based on false or misleading internal information — presents a significant challenge for the application of IndAI in complex settings. In Industrial 5.0 scenarios, such errors can lead to production disruptions or safety risks, particularly in environments that require high precision, such as automated manufacturing and human-robot collaboration. For the interpretability and explainability of large and small IndAI models, several topics, including multi-modal perception fusion, causal reasoning, real-time error detection and correction, and adversarial training, should be studied. The mainstream IndAI models such as DJI RoboMaster, CloudMinds, Baidu Apollo, Xiaomi CyberDog, and iFlytek's Service Robots are being actively developed [131]. These systems rely heavily on robust network infrastructure to function effectively.

b) Embodied Learning With Data Enhancement: A major challenge in IndAI is the scarcity of diverse, high-quality training data, especially in specialized industrial contexts. Future efforts should prioritize using synthetic data generation, data augmentation, and transfer learning to bridge this gap. Researchers can reduce reliance on costly real-world data collection by creating large-scale, domain-specific datasets and realistic synthetic data. Additionally, enhancing training in simulated environments with techniques like domain randomization, adversarial training, and context-aware learning can help mitigate Sim2Real performance gaps, ensuring models can seamlessly transition to real-world applications.

c) collaborative Embodied Learning Between Large and Small Models: Leveraging the powerful capability of large models to deeply mine data value, this approach breaks down information silos between processes and fills the knowledge gaps across multiple scenarios. It collaborates with small models to make precise decisions in specialized scenarios. Through a dual-system collaborative inference mechanism involving both large and small models, it enables end-to-end inference across different scenarios, processes, and workflows. This includes applications in business management, production management, monitoring and control, sensing and actuation, and industrial equipment, thereby breaking down information silos and bridging knowledge gaps.

2) Link Layer: Another critical component in Industry 5.0 are the links, ensuring the seamless and efficient communication between devices, sensors, and systems. This layer plays a pivotal role in enabling real-time data processing and communication, which are essential for the smooth operation of entities in industrial settings.

a) Networked Embodied Cognition and Enactivism: The performance of IndAI systems, particularly those involving real-time data processing and communication, is highly dependent on network bandwidth. Adequate bandwidth ensures low latency and high reliability, which are essential for the seamless operation of robots and sensors in both industrial and general settings. Insufficient bandwidth can lead to delays and data loss, significantly degrading the performance and safety of these systems. One of the significant research challenges and limitations faced by the multi-agent community has been achieving higher levels of intelligence in the networked environment. While considerable attention has been given to topics such as emergent hierarchies, group size formation, and migration patterns, less focus has been placed on integrating advanced cognitive and enactive processes like thinking, reasoning, and language into networked MASs. These higher-level cognitive and enactive functions have typically not been central to the research agenda, highlighting the need for more comprehensive approaches that combine population dynamics with sophisticated cognitive mechanisms.

b) Embodied Software and Hardware Multifunctionality: The emergence of SI in MI system not only requires intelligent algorithms but also depends on multifunctional software and hardware technologies. To achieve this, a comprehensive and open unmanned cluster software and hardware technology stack is needed to integrate relevant swarm intelligence algorithms and technical components. This integration aims to realize the group-level perception, cognition, decision-making, and action control loop. Although machine intelligence varies greatly in shape, size, and movement functions, they are essentially all intelligent embedded systems that need to follow the design methods of embedded systems. This includes integrating sensors, servos, embedded intelligent models, and other components for comprehensive design and verification.

3) Application Layer: The application layer in Industry 5.0 focuses on the practical implementation and deployment of intelligent systems in real-world scenarios.

a) Adaptive Long-Horizon Embodied Task Planning: IndAI systems tackle long-horizon tasks in highly complex and dynamic environments. However, maintaining context and goal awareness over extended periods remains a significant challenge. Future research should focus on developing hierarchical planning and reinforcement learning frameworks that integrate memory modules and context-aware strategies to enhance decision-making capabilities. This approach would enable these systems to autonomously adapt their strategies based on real-time feedback, allowing them to efficiently execute multi-step, long-duration tasks in unpredictable industrial scenarios, thereby minimizing the need for human intervention and aligning with the human-centric vision of Industry 5.0.

b) LLM-based Embodied MASs in Multimodal Environments: Most previous work on LLM-based MASs has focused on text-based environments, excelling in processing and generating textual content. However, in multimodal environments, agents will interact with a variety of sensory inputs and produce multiple types of outputs, such as images, audio, video, and physical actions, which represents a significant gap. Integrating LLMs into multimodal environments presents additional challenges, such as handling different types of data and enabling agents to understand each other and respond not just to textual information but also to other forms of input.

VII. Conclusions

In this work, the development of smart and human-centric MI under Industry 5.0 has been reviewed. As the next frontier in industrialization, the core of Industry 5.0 shifts from purely technology-driven automation to a human-centric approach. The emergence of embodied intelligence, cloud computing, IoT, big data, reinforcement learning, has made the realization of IndAI in the framework of Industry 5.0 not only feasible but also transformative. These technologies enable the fusion of the physical and virtual worlds through MAS, and the integration of IndAI with human intelligence for the augmented intelligence. Then, the key technologies and their potential applications in supporting single-agent, multi-agent, and swarm-agent embodied IndAI are introduced. These include smart perception, control, scheduling, communication, MLLMs, and collaborative training environments. Each of these technologies plays a crucial role in enhancing the capabilities of embodied agents, making them more adaptive, responsive, and intelligent. Finally, we have identified and discussed several open challenges and opportunities in the application of Industry 5.0 to smart MI. These challenges include ensuring data security and privacy, managing the complexity of integrated systems, and addressing the ethical and social implications of augmented intelligence. Despite these challenges, the opportunities are vast, ranging from improved productivity and sustainability to enhanced worker well-being and innovation.

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